

MLE properties, sufficient statistics

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MLE existence example

Example: consider n IID observations from uniform distribution with PDF:

$$f(x) = \begin{cases} \frac{1}{\theta} & x \in (0, \theta) \\ 0 & \text{otherwise} \end{cases}$$

Discuss the existence of MLE estimator for parameter θ .

$$L(\theta) = \prod_{i=1}^n \frac{1}{\theta} = \frac{1}{\theta^n} = \theta^{-n} \quad x \in (0; \theta)$$

$$l(\theta) = -n \ln(\theta) \quad \frac{\partial l}{\partial \theta} = -\frac{n}{\theta} \quad \rightarrow \max_{x_i} \{x_i\} = \hat{\theta}$$

MLE necessary

MLE uniqueness example

Example: consider n IID observations from uniform distribution with PDF:

$$f(x) = \begin{cases} 1 & x \in \langle \theta, \theta + 1 \rangle, \\ 0 & \text{otherwise.} \end{cases}$$

Use your knowledge of likelihood to estimate θ .

$$L(\theta) = \prod_{i=1}^n 1 = 1 \quad \theta \in \langle \theta; \theta+1 \rangle$$

$$\theta \leq \min \{x_i\}$$

$$\forall i, x_i \in \langle \theta; \theta+1 \rangle$$

$$\theta \geq \max \{x_i\} - 1$$

MLE invariance

Theorem

if $\hat{\theta}$ is the MLE of θ and if $g(\theta)$ is one-to-one function, then $g(\hat{\theta})$ is MLE for $g(\theta)$

Example: Find a MLE estimator for limits of an interval $\langle \mu - 3\sigma; \mu + 3\sigma \rangle$.

Regularity conditions

Let $\{f(\mathbf{x}, \theta), \theta \in \Theta\}$ be a system of PDFs (PMFs). This system is regular if the following conditions are met:

- 1) Parametric space Θ is nonempty, open set.
- 2) A set $M = \{\mathbf{x} : f(\mathbf{x}, \theta) > 0\}$ does not depend on the value of θ .
- 3) For each $\mathbf{x} \in M$ a finite partial derivative $f'(\mathbf{x}, \theta) = \frac{\partial f(\mathbf{x}, \theta)}{\partial \theta}$ exists.
- 4) For all $\theta \in \Theta$ holds $\int_M f'(\mathbf{x}, \theta) d\mathbf{x} = 0$
- 5) Integral:

$$J_n(\theta) = \int_M \left[\frac{f'(\mathbf{x}, \theta)}{f(\mathbf{x}, \theta)} \right]^2 f(\mathbf{x}, \theta) d\mathbf{x}$$

is finite and positive

Consistent estimator

Definition

A sequence of estimators that converges in probability to the unknown value of the parameter being estimated, as $n \rightarrow \infty$, is called a consistent sequence of estimators.

Theorem

MLEs satisfying regularity conditions form a consistent sequence of estimators.

Unbiased estimator

Definition (Nestranný odhad)

Let statistic $\delta(X)$ be an estimator of a function $g(\theta)$. We say that $\delta(X)$ is unbiased if $E_{\theta}[\delta(X)] = g(\theta)$ for all values of θ .

Theorem

Arithmetic mean $\bar{X}$ is an unbiased estimator of parameter μ in normal distribution.

$$E(\bar{X}) \stackrel{?}{=} \mu \quad E(X_i) = \mu \quad ; X_i \sim N(\mu, \sigma^2)$$

$$E\left(\frac{1}{n} \sum_{i=1}^n \cancel{X_i}\right) = \frac{1}{n} E\left(\sum_{i=1}^n X_i\right) = \frac{1}{n} \sum_{i=1}^n E(X_i) = \frac{1}{n} \sum_{i=1}^n \mu = \frac{1}{n} \cdot n\mu = \mu$$

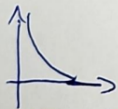
Jensen's inequality

Theorem

Let g be a strictly convex (concave) function and let X be a random variable with finite mean. Then $E[g(X)] > g[E(X)]$ ($E[g(X)] < g[E(X)]$).

Example: Discuss whether an MLE for λ in exponential distribution is unbiased. If the estimate is biased use Jensen's inequality to determine a "direction" of bias.
bias illustration in Python

$$f(x) = \begin{cases} \lambda \cdot e^{-\lambda x} & x \geq 0 \\ 0 & \text{sonst} \end{cases} \quad \text{MLE } \lambda = \hat{\lambda} = \frac{1}{\bar{x}} \quad E(X_i) = \frac{1}{\lambda}$$



$$E(\hat{\lambda}) = E\left(\frac{1}{\bar{x}}\right) \neq \frac{1}{E(\bar{x})} = \frac{1}{\frac{1}{n} \sum_{i=1}^n \frac{1}{\lambda}} = \lambda$$

$$E(g(x)) \geq g(E(x)) \quad g = \frac{1}{x}$$

$$\text{MTTF} = E(X_i) = \frac{1}{\lambda}$$

$$\hat{\text{MTTF}} = \bar{x} \quad E(\bar{x}) = \frac{1}{\lambda} = \text{MTTF}$$

Sufficient statistic

Definition (Postačující statistika)

Let $\mathbf{X}$ be a random vector of IID random variables. T is called a sufficient statistic for parameter θ , if conditional joint PDF (PMF) of $\mathbf{X}$ given $T = t$ does not depend on θ

Theorem

Let $T(\mathbf{X})$ be a sufficient statistic for the parameter θ and let $g(T(\mathbf{X}))$ be a measurable one-to-one function, then $g(T(\mathbf{X}))$ remains sufficient statistic for the parameter θ .

Corollary of sufficiency

Suppose that statisticians A and B are trying to estimate the same parameter θ . Statistician A has access to all observed values on the contrary, statistician B can use only a statistic of observed values. Generally speaking statistician A has a significant advantage for estimation of θ , however if sufficient statistic for the parameter θ exists (and statistician B uses sufficient statistic) statistician A can not create better estimate for θ .

Sufficient statistics summarize all necessary information for the purpose of estimating θ without loss.

If sufficient statistic for parameter θ is observed, knowing all observed values gives no advantage for estimating θ .

Sufficiency for a parameter vector

Concept of sufficiency can be generalized for a vector of parameters. If the vector of statistics has the sufficient property, the statistics are called *Jointly sufficient*.

If the dimension of the vector can not be reduced, and the statistics are jointly sufficient, then the statistics are called *Minimal sufficient statistics*.

Factorization criterion

Theorem

Let $r_1, \dots, r_k$ be functions of n real variables. The statistics $T_i = r_i(\mathbf{X}), i = 1, \dots, k$, are jointly sufficient statistics for θ if and only if the joint PMF or the joint PDF $f_n(\mathbf{x}, \theta)$ can be factored as follows for all values of $\mathbf{x} \in \mathbb{R}^n$ and all values of $\theta \in \Theta$:

$$f_n(\mathbf{x}, \theta) = u(\mathbf{x})v[r_1(\mathbf{x}), \dots, r_k(\mathbf{x}), \theta]. \quad (1)$$

Here the functions u and v are nonnegative, the function u may depend on $\mathbf{x}$ but does not depend on θ , and the function v will depend on θ but depends on $\mathbf{x}$ only through the k functions $r_1(\mathbf{x}), \dots, r_k(\mathbf{x})$.

Exponential family

Definition - general formula

If the PDF or PMF can be expressed as follows, we say that the random variable belongs to the exponential family.

$$f(x, \theta) = h(x) \exp(\eta(\theta) \cdot T(x) - A(\theta))$$

where:

$h(x)$ is a function of the data x that does not depend on the parameter θ .

$\eta(\theta)$ is the natural parameter.

$T(x)$ is a function of x .

$A(\theta)$ is the log-partition function, ensuring that the PDF (PMF) integrates (sums) to 1

Definition - location, scale parametrization

If the PDF or PMF can be expressed as follows, we say that the random variable belongs to the exponential family.

$$f(x, \theta, \phi) = h(x) \exp\left\{\frac{1}{\phi} [\eta(\theta) \cdot x - B(\eta(\theta))] + c(x, \phi)\right\}$$

where:

$h(x)$ is a function of the data x that does not depend on the parameters θ or ϕ .

ϕ is a dispersion parameter

$\eta(\theta)$ is the natural parameter.

$c(x, \phi)$ is a function of x and a dispersion parameter ϕ .

$B(\eta(\theta))$ is strictly convex twice differentiable function

Theorem

If a probability distribution belongs to the exponential family, then minimal sufficient statistics exist and can be found using the factorization criterion.

Theorem

Moments of the distributions of the exponential family can be derived as: $E[X] = \frac{d}{d\eta(\theta)} B(\eta(\theta))$, $Var[X] = \phi \frac{d^2}{d[\eta(\theta)]^2} B(\eta(\theta))$

Example: Find sufficient statistic for a probability parameter π for N IID binomially distributed random variable with fixed (and known parameter n). Show that this probability distribution belongs to exponential family. Derive its expectation and variance.

$X_1, \dots, X_N \sim \text{i.i.d.}$ n - sample

$X_i \sim \text{Bi}(n, \pi)$ $p(x) = \binom{n}{x} \pi^x (1-\pi)^{n-x}$ $x \in \{0, 1, 2, \dots, n\}$

$e^{\ln(x)} = x$

$p_{\pi}(x_1, \dots, x_N; \pi) = \prod_{i=1}^N \binom{n}{x_i} \pi^{x_i} (1-\pi)^{n-x_i} = \left[\prod_{i=1}^N \binom{n}{x_i} \right] \cdot \pi^{\sum_{i=1}^N x_i} \cdot (1-\pi)^{\sum_{i=1}^N (n-x_i)} =$

$= h(x_i) \cdot \exp \left\{ \sum_{i=1}^N x_i \ln \pi \right\} \cdot \exp \left\{ \sum_{i=1}^N (n-x_i) \ln (1-\pi) \right\} =$

$= h(x_i) \cdot \exp \left\{ \sum_{i=1}^N x_i \ln(\pi) + Nn \cdot \ln(1-\pi) - \sum_{i=1}^N x_i \cdot \ln(1-\pi) \right\}$

$\Rightarrow T(X) = \sum_{i=1}^N x_i$

$p(x) = \binom{n}{x} \cdot \exp \left\{ x \ln(\pi) + n \cdot \ln(1-\pi) - x \cdot \ln(1-\pi) \right\} =$

$= \binom{n}{x} \cdot \exp \left\{ x \cdot \ln \left(\frac{\pi}{1-\pi} \right) + n \cdot \ln(1-\pi) \right\}$

$= \binom{n}{x} \cdot \exp \left\{ x \cdot \eta - n \cdot \ln(1+e^\eta) \right\}$

$\left| \begin{aligned} \eta &= \ln \left(\frac{\pi}{1-\pi} \right) \\ e^\eta &= \frac{\pi}{1-\pi} \rightarrow \pi = \frac{e^\eta}{1+e^\eta} \\ \ln \left(1 - \frac{e^\eta}{1+e^\eta} \right) &= \ln \left(\frac{1}{1+e^\eta} \right) = -\ln(1+e^\eta) \end{aligned} \right|$

$E(X) = \frac{d}{d\eta} \cdot n \cdot \ln(1+e^\eta) = n \cdot \frac{e^\eta}{1+e^\eta} = n\pi$

$\text{Var}(X) = \frac{d^2}{d\eta^2} n \cdot \ln(1+e^\eta) = n \cdot \frac{e^\eta(1+e^\eta) - e^{2\eta}}{(1+e^\eta)^2} = n \cdot \frac{e^\eta}{(1+e^\eta)^2} = n \cdot \frac{e^\eta}{1+e^\eta} \cdot \frac{1}{1+e^\eta} = n\pi(1-\pi)$